

CSC10006 – Introduction to Database

Nested query

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1 Objectives

After completing this lab, students will be able to:

- Understand the concepts of nested queries, hierarchical nested queries, and correlated subqueries
- Write hierarchical nested queries
- Use the operators IN, NOT IN, ALL, ANY, and SOME
- Use the operators EXISTS and NOT EXISTS
- Apply hierarchical nested queries
- Apply correlated subqueries
- Use nested queries in combination with grouping

2 Detailed Instructions

2.1 Introduction to Nested Queries

Definition: A **nested query** (subquery) is a query that contains another query within its statement.

Syntax:

```
SELECT A
FROM X
WHERE ... (SELECT B FROM Y WHERE ...) ...
```

Based on the characteristics of subqueries, nested queries can be classified into two main types:

Hierarchical Nested Queries: A hierarchical nested query is a query in which the subquery is independent of the outer (parent) query

```
SELECT A
FROM X
WHERE ... (SELECT B, C FROM Y) ...
```

In this example, the subquery `SELECT B, C FROM Y` does **not** reference any component of the outer query. Therefore, it is classified as a **hierarchical nested query**.

Correlated Subqueries: A correlated subquery is a query in which the subquery depends on the outer query

```
SELECT A
FROM X
WHERE ... (SELECT B, C FROM Y WHERE B = X.A) ...
```

In this example, the subquery `SELECT B, C FROM Y WHERE B = X.A` references a column from the outer query through the comparison `B = X.A`. Therefore, this is a correlated subquery.

2.2 Placement of Subqueries in the Outer Query:

A subquery can appear in different parts of the outer query. Common positions include the `SELECT`, `FROM`, and most commonly, the `WHERE` clause.

2.2.1 Subquery in the `SELECT` Clause

The result of the subquery is treated as a **single attribute value**.

Example 1: For each department, retrieve the department name and the number of teachers in that department.

```
SELECT BM.TENBM, ( SELECT COUNT(*)
                    FROM GIAOVIEN GV
                    WHERE GV.MABM = BM.MABM)
FROM BOMON BM
```

2.2.2 Subquery in the `FROM` Clause

The result of the subquery is treated as a **derived table**, which can be queried like a regular table.

Example 2: Retrieve the full name and salary of teachers in the HTTT department

```

SELECT T.HOTEN, T.LNG
FROM (    SELECT MAGV, HOTEN, LUONG as LNG
        FROM GIAOVIEN
        WHERE MABM= 'HTTT') as T

```

2.2.3 Subquery in the WHERE Clause

The result of the subquery is used as part of a **conditional expression**

Example 3: Retrieve teachers whose salary is greater than the salary of the teacher with MAGV = '001'

```

SELECT *
FROM GIAOVIEN
WHERE LUONG > (SELECT LUONG
               FROM GIAOVIEN
               WHERE MAGV= '001')

```

2.3 Hierarchical Nested Queries with the IN Operator

The **IN** operator is used to check whether a value belongs to a given set of values. This set can be the result of a subquery or an explicitly specified list of values (see Topic 03 – Basic Queries).

Syntax: <attribute> **IN** <set>

```

<attribute> IN (    SELECT ...
                  FROM ...
                  WHERE ...)

```

Or:

```

<attribute> IN (value1, value2, ..., valueN)

```

The condition evaluates to **TRUE** if the attribute value appears in the set; otherwise, it evaluates to **FALSE**.

Example 4: Retrieve the names of teachers who do not have any relatives.

```
SELECT HOTEN
FROM GIAOVIEN
WHERE MAGV NOT IN (SELECT MAGV FROM NGUOITHAN)
```

Example 5: Retrieve teachers who participate in at least one project

```
SELECT *
FROM GIAOVIEN
WHERE MAGV IN (SELECT MAGV FROM THAMGIADT)
```

2.4 Hierarchical Nested Queries with the ALL Operator

The **ALL** operator is used together with comparison operators such as **>**, **<**, **>=**, **<=**, **!=**, etc.

Syntax: <attribute> [**>** | **<** | **>=** | **<=** | **!=**] **ALL** <set>

This condition evaluates to **TRUE** if the attribute value satisfies the comparison **with all elements** in the set

Example 6: Retrieve teachers who have the lowest salary.

```
SELECT *
FROM GIAOVIEN
WHERE LUONG <= ALL (SELECT LUONG FROM GIAOVIEN)
```

Example 7: Retrieve teachers whose salary is higher than all teachers in the HTTT department.

```
SELECT *
FROM GIAOVIEN
WHERE LUONG >=ALL(SELECT LUONG
                    FROM GIAOVIEN
                    WHERE MABM= 'HTTT')
```

Example 8: Retrieve the department code(s) (MABM) with the largest number of teachers.

```
SELECT MABM
FROM GIAOVIEN
GROUP BY MAMB
HAVING COUNT(*) >= ALL( SELECT COUNT(*)
                        FROM GIAOVIEN
                        GROUP BY MABM)
```

Example 9: Retrieve the names of teachers who do not have any relatives (using ALL instead of NOT IN).

```
SELECT HOTEN
FROM GIAOVIEN
GROUP BY MAGV != ALL ( SELECT MAGV
                      FROM NGUOITHAN)
```

2.5 Hierarchical Nested Queries with ANY and SOME

The syntax for using **ANY** and **SOME** is similar to that of **ALL**.

- **ALL**: the condition must be satisfied for **all values** in the set.
- **ANY / SOME**: the condition must be satisfied for **at least one value** in the set.

Example 10: Retrieve the names of teachers who participate in at least one project (using $\equiv$ ANY instead of IN).

```
SELECT HOTEN
FROM GIAOVIEN
WHERE MAGV = ANY (SELECT MAGV
                  FROM THAMGIADT)
```

Explanation:

- = ANY (subquery) means the value matches **at least one** value returned by the subquery.
- This query is logically equivalent to using IN.

2.6 Correlated Subqueries with EXISTS

EXISTS is commonly used in conditional expressions

```
... EXISTS ( SELECT ...
            FROM ...
            WHERE ... )
```

- The EXISTS condition evaluates to TRUE if the subquery returns one or more rows.
- If the subquery returns no rows, the condition evaluates to FALSE
- EXISTS is most commonly used in correlated subqueries.
- Unlike IN, EXISTS only checks for the existence of rows, not actual values, and is often more efficient for large datasets

Example 11: Retrieve teachers who participate in at least one project.

```
SELECT HOTEN
FROM GIAOVIEN GV
WHERE EXISTS ( SELECT *
              FROM THAMGIADT TG
              WHERE TG.MAGV = GV.MAGV)
```

Example 12: Retrieve teachers who do not have any relatives

```
SELECT HOTEN
FROM GIAOVIEN GV
WHERE NOT EXISTS ( SELECT *
                    FROM NGUOITHAN NT
                    WHERE NT.MAGV = GV.MAGV)
```

Example 13: Retrieve teachers who do not have any relatives

```
SELECT HOTEN
FROM GIAOVIEN GV
WHERE NOT EXISTS ( SELECT *
                    FROM NGUOITHAN NT
                    WHERE NT.MAGV = GV.MAGV)
```

2.7 Additional Examples of Correlated Subqueries

Example 14: Retrieve teachers whose salary is higher than the average salary of the department in which they work.

```
SELECT HOTEN
FROM GIAOVIEN GV1
WHERE GV1.LUONG > ( SELECT AVG (GV2.LUONG)
                    FROM GIAOVIEN GV2
                    WHERE GV2.MABM = GV1.MABM)
```

Example 15: Retrieve teachers who have the highest salary.

```
SELECT HOTEN
FROM GIAOVIEN GV1
WHERE ( SELECT COUNT(*)
        FROM GIAOVIEN GV2
        WHERE GV2.LUONG > GV1.LUONG) = 0
```

Example 16: Retrieve projects in which teacher '001' does not participate

Solution 1: Using NOT EXISTS

```
SELECT MADT
FROM DETAI DT
WHERE NOT EXISTS ( SELECT *
                   FROM THAMGIADT TG
                   WHERE TG.MADT = DT.MADT AND MAGV = '001')
```

Solution 2: Using NOT IN

```
SELECT MADT
FROM DETAI
WHERE MADT NOT IN ( SELECT MADT
                   FROM THAMGIADT
                   WHERE MAGV = '001')
```

2.8 A Query Can Be Solved in Multiple Ways

Example 17: Retrieve the names of teachers who have an academic supervisory role over other teachers

Using EXISTS:

```
SELECT GV1.HOTEN
FROM GIAOVIEN GV1
WHERE EXISTS ( SELECT *
                FROM GIAOVIEN GV2
                WHERE GV2.GVQLCM = GV1.MAGV)
```

Using IN:

```
SELECT HOTEN
FROM GIAOVIEN
WHERE MAGV IN ( SELECT GVQLCM
                FROM GIAOVIEN)
```

Using join:

```
SELECT DISTINCT GV2.HOTEN
FROM GIAOVIEN GV1, GIAOVIEN GV2
WHERE GV1.GVQLCM = GV2.MAGV
```

2.9 Nested Queries with Aggregation and Grouping

Example 18: Retrieve teachers who have the highest salary.

```
SELECT MABM
FROM GIAOVIEN
WHERE LUONG =(SELECT MAX(LUONG)
               FROM GIAOVIEN)
OR
SELECT MABM
FROM GIAOVIEN
WHERE LUONG >= ALL (SELECT LUONG
                    FROM GIAOVIEN)
```

Example 19: Retrieve the department code(s) (MABM) with the highest number of teachers

```
SELECT MABM
FROM GIAOVIEN
GROUP BY MABM
HAVING COUNT(*) >= ALL ( SELECT COUNT(*)
                           FROM GIAOVIEN GV
                           GROUP BY GV.MABM)

OR

SELECT MABM
FROM GIAOVIEN
GROUP BY MABM
HAVING COUNT(*) = ( SELECT MAX (SLGV)
                    FROM ( SELECT COUNT(*) AS SLGV
                           FROM GIAOVIEN
                           GROUP BY MABM) AS T )
```

Example 20: Retrieve the department name, the department head's name, and the number of teachers for the department(s) with the highest number of teachers

```
SELECT BM.TENBM, TBM.HOTEN, COUNT(*) AS SLGV
FROM GIAOVIEN GV, BOMON BM, GIAOVIEN TBM
WHERE GV.MABM = BM.MABM AND BM.TRUONGBM=TBM.MAGV
GROUP BY GV.MABM, TBM.HOTEN
HAVING COUNT(*) >= ALL ( SELECT COUNT(*)
                           FROM GIAOVIEN
                           GROUP BY MABM)
```

Example 21: Retrieve teachers whose salary is higher than the average salary of teachers in the Information Systems department, but who do not belong to the Information Systems department.

```
SELECT GV.HOTEN
FROM GIAOVIEN GV
WHERE LUONG > ( SELECT AVG(GV.LUONG)
                FROM GIAOVIEN GV, BOMON BM
                WHERE GV.MABM=BM.MABM AND
                      BM.TENBM = 'Hệ thống thông tin')
```

Example 22: Retrieve the name(s) of the project(s) with the highest number of participating teachers

```
SELECT MAX (MADT)
FROM DETAI
WHERE LUONG > ( SELECT MADT, COUNT (DISTINCT MAGV) AS SL
                FROM THAMGIADT
                GROUP BY MADT )
```

3. In-class exercises

Requirement: Write SQL queries **Q35, Q37, ..., Q57** for the “Quản lý đề tài” exercise

Duration: 2 hour.

Q35. Retrieve the highest salary among lecturers.

Q36. Retrieve teachers who have the highest salary.

Q37. Retrieve the highest salary in the “HTTT” department.

Q38. Retrieve the name of the oldest teacher in the Information Systems department.

Q39. Retrieve the name of the youngest teacher in the Faculty of Information Technology.

Q40. Retrieve the teacher’s name and the faculty name of the teacher with the highest salary.

Q41. Retrieve teachers who have the highest salary within their own department.

- Q42.** Retrieve the names of projects that the teacher “Nguyễn Hoài An” has not participated in.
- Q43.** Retrieve the projects that the teacher “Nguyễn Hoài An” has not participated in.
Output the project name and the name of the project leader.
- Q44.** Retrieve the names of teachers in the Faculty of Information Technology who have not participated in any project.
- Q45.** Find teachers who do not participate in any project.
- Q46.** Retrieve teachers whose salary is higher than the salary of the teacher “Nguyễn Hoài An”.
- Q47.** Find department heads who participate in at least one project.
- Q48.** Find teachers who have the same name and gender as another teacher in the same department.
- Q49.** Find teachers whose salary is higher than the salary of at least one teacher in the “Công nghệ phần mềm” department.
- Q50.** Find teachers whose salary is higher than the salary of all teachers in the “Hệ thống thông tin” department.
- Q51.** Retrieve the name of the faculty that has the largest number of teachers.
- Q52.** Retrieve the full name of the teacher who is the leader of the largest number of projects.
- Q53.** Retrieve the department code that has the largest number of teachers.
- Q54.** Retrieve the teacher’s name and department name of the teacher who participates in the largest number of projects.
- Q55.** Retrieve the name of the teacher in the HTTT department who participates in the largest number of projects.
- Q56.** Retrieve the teacher’s name and department name of the teacher who has the largest number of relatives.
- Q57.** Retrieve the name of the department head who is the leader of the largest number of projects.